# Practice workbook · Buses and the fetch-decode-execute cycle

What the address, data and control buses carry and in which direction, the effect of bus width, how the program counter moves on, and putting the cycle's stages in order.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-04

# 1. Question 2 · part c.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.ii only.

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**2** Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

**(2)**

1 .....

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2 .....

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(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

**(1)**

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

**(2)**

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# 1. Question 2 · part c.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.ii only.

(c) The CPU performs the fetch-decode-execute cycle.

(i) Give **three** factors that affect the performance of the CPU.

(3)

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2 .....

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3 .....

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(ii) Describe how the address bus is used during the fetch-decode-execute cycle.

(2)

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(d) Classical computers store data using bits that exist in one of two binary states, 1 or 0.

Quantum computers store data in a superposition of states.

Give **two other** characteristics of quantum computers.

(2)

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2 .....

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**(Total for Question 2 = 12 marks)**

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## 1. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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## 2. Question 3 · part a · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

**3** The central processing unit (CPU) and other hardware components work together with the operating system to run programs and store data.

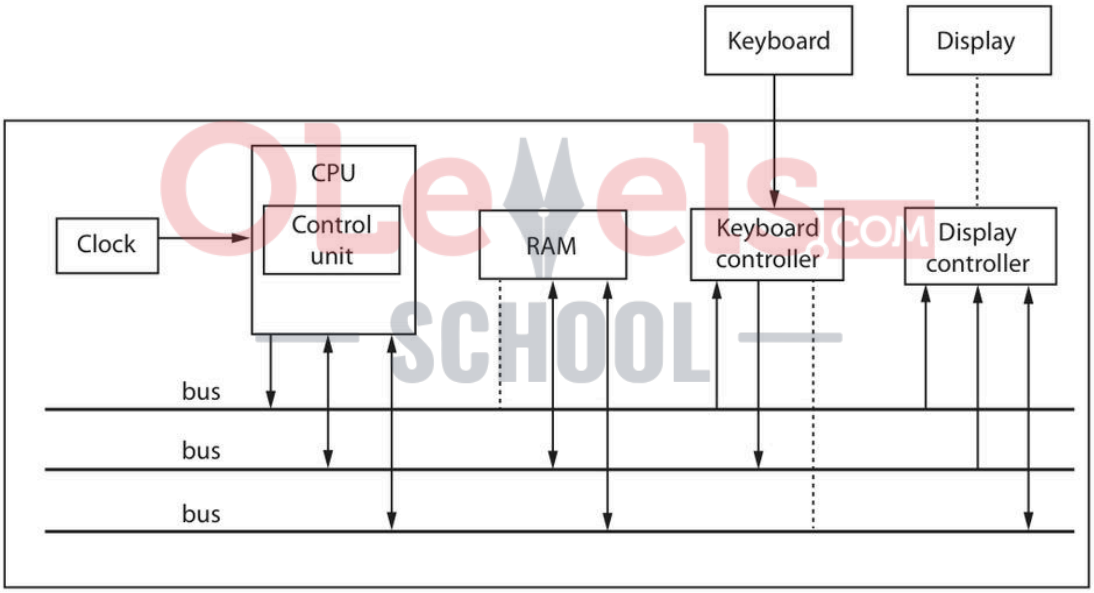
(a) Here is a partially completed diagram showing the hardware components used in the fetch-decode-execute cycle.

The function of the I/O controller is to manage communication between the CPU and input/output devices.

Complete the diagram by:

- labelling the buses
- adding directional arrows to the dotted lines to show the flow of communication.

**(6)**




(b) The CPU includes an area of cache memory.

Increasing the size of the cache will improve the performance of the CPU.

Explain how increasing the size of the cache improves the performance.

**(2)**

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## 2. Question 3 · part a · 6 marks · Working and answer

Working and answer

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## 3. Question 4 · part d.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.iii only.

**4** Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

**(1)**

(b) Explain **one** reason why a simulation may require a powerful processor.

**(2)**

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

**(3)**

| R | S | W | NOT S | R AND W | (NOT S) OR (R AND W) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

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## 3. Question 4 · part d.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.iii only.

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(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

(2)

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

(2)

1 .....

2 .....

(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

(2)

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

(2)

| Change made | Effect |
|---|---|
| Increasing the width of the data bus | |
| Increasing the width of the address bus | |

**(Total for Question 4 = 14 marks)**

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## 3. Question 4 · part d.iii · 2 marks · Working and answer

Working and answer

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# 4. Question 4 · part d.iv · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.iv only.

**4** Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

**(1)**

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

**(2)**

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

**(3)**

| R | S | W | NOT S | R AND W | (NOT S) OR (R AND W) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

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# 4. Question 4 · part d.iv · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.iv only.

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**(2)**

| Change made | Effect |
|---|---|
| Increasing the width of the data bus | |
| Increasing the width of the address bus | |

**(Total for Question 4 = 14 marks)**

## 4. Question 4 · part d.iv · 2 marks · Working and answer

Working and answer

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# 5. Question 5 · part b.i · 2 marks

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Further practice · Attempt part b.i only.

**5** Computers can be general-purpose or specific-purpose machines.

(a) The central processing unit (CPU) is responsible for executing a program.

**Figure 4** shows some of the hardware components of a computer system.

Complete the table by matching the component to the letters shown in **Figure 4**.

**(5)**

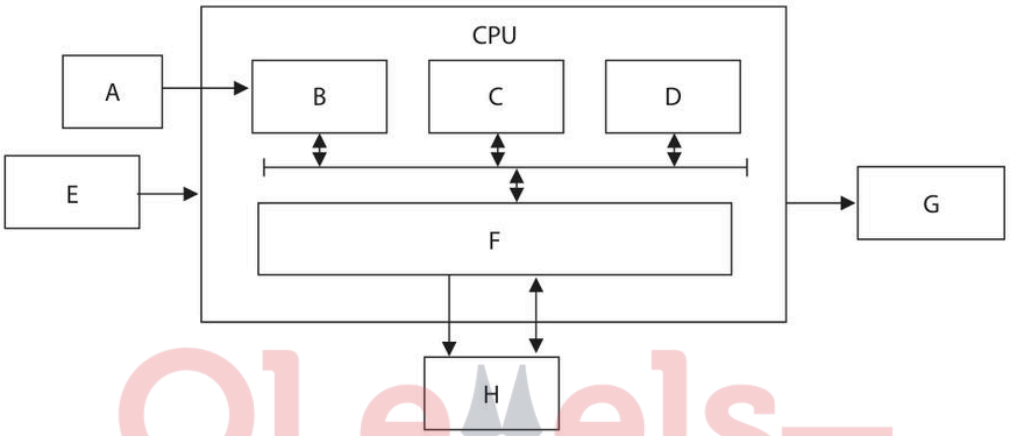



**Figure 4**

| Component | Letter |
|---|---|
| Control unit | |
| Output device | |
| Registers | C |
| Main memory | |
| Cache | F |
| Input device | |
| Arithmetic/logic unit (ALU) | |
| Clock | A |

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# 5. Question 5 · part b.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

**(2)**

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

**(1)**

- ☒ **A** Memory delete signal
- ☒ **B** Memory read signal
- ☒ **C** Memory store signal
- ☒ **D** Memory write signal

(iii) Clock speed is the number of pulses the CPU's clock generates per second.

State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

**(1)**

(c) An embedded system forms part of a larger system, device or machine.

(i) Give **two** features of an embedded system.

**(2)**

1 .....

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## 5. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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## 6. Question 5 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

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(a) The central processing unit (CPU) is responsible for executing a program.

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Complete the table by matching the component to the letters shown in **Figure 4**.

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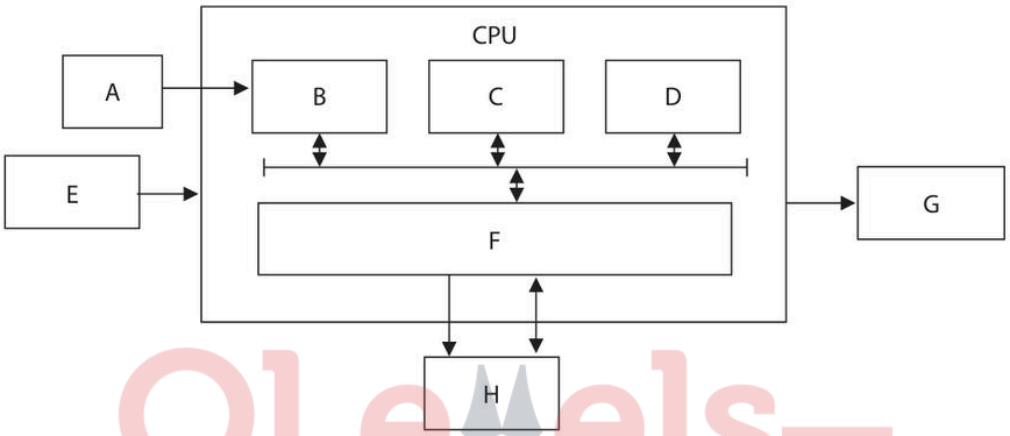



**Figure 4**

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|---|---|
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| Output device | |
| Registers | C |
| Main memory | |
| Cache | F |
| Input device | |
| Arithmetic/logic unit (ALU) | |
| Clock | A |

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# 6. Question 5 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

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## 6. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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# 7. Question 2 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

**2** Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

**(1)**

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(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

**(3)**

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# 7. Question 2 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

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(b) Computers use different types of storage.

(i) Explain **one** reason why computers need both RAM and ROM.

**(2)**

(ii) Describe how data is read from a magnetic hard drive.

**(3)**

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(c) Computers perform the fetch-decode-execute cycle.

Complete the table by writing the numbers 1 to 4 in the sequence number column to put each stage in the correct order.

| Stage | Sequence number |
|---|---|
| The instruction is placed onto the data bus | |
| The Arithmetic and Logic Unit (ALU) performs a calculation | |
| The address of the next instruction is placed onto the address bus | |
| The instruction is decoded by the control unit | |

**(4)**

## 7. Question 2 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

(d) Identify which **one** of these buses is unidirectional.

(1)

- ☒ **A** Address bus
- ☒ **B** Control bus
- ☒ **C** System bus
- ☒ **D** Data bus

**(Total for Question 2 = 14 marks)**

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Working and answer

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## 8. Question 2 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.

**2** Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

**(1)**

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

**(3)**

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## 8. Question 2 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.

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**(3)**

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| Stage | Sequence number |
|---|---|
| The instruction is placed onto the data bus | |
| The Arithmetic and Logic Unit (ALU) performs a calculation | |
| The address of the next instruction is placed onto the address bus | |
| The instruction is decoded by the control unit | |

**(4)**

## 8. Question 2 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.

(d) Identify which **one** of these buses is unidirectional.

(1)

- ☒ **A** Address bus
- ☒ **B** Control bus
- ☒ **C** System bus
- ☒ **D** Data bus

**(Total for Question 2 = 14 marks)**

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Working and answer

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